UML Diagrams

serial-app

Black Sphere Industries

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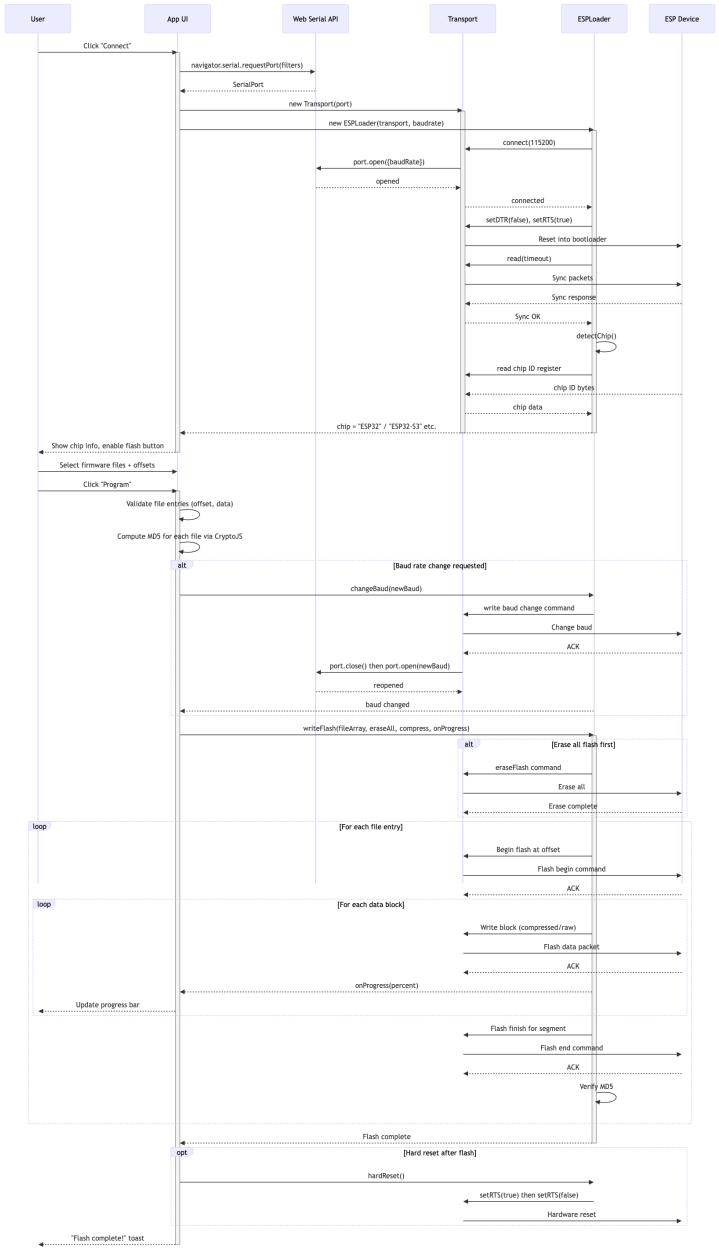
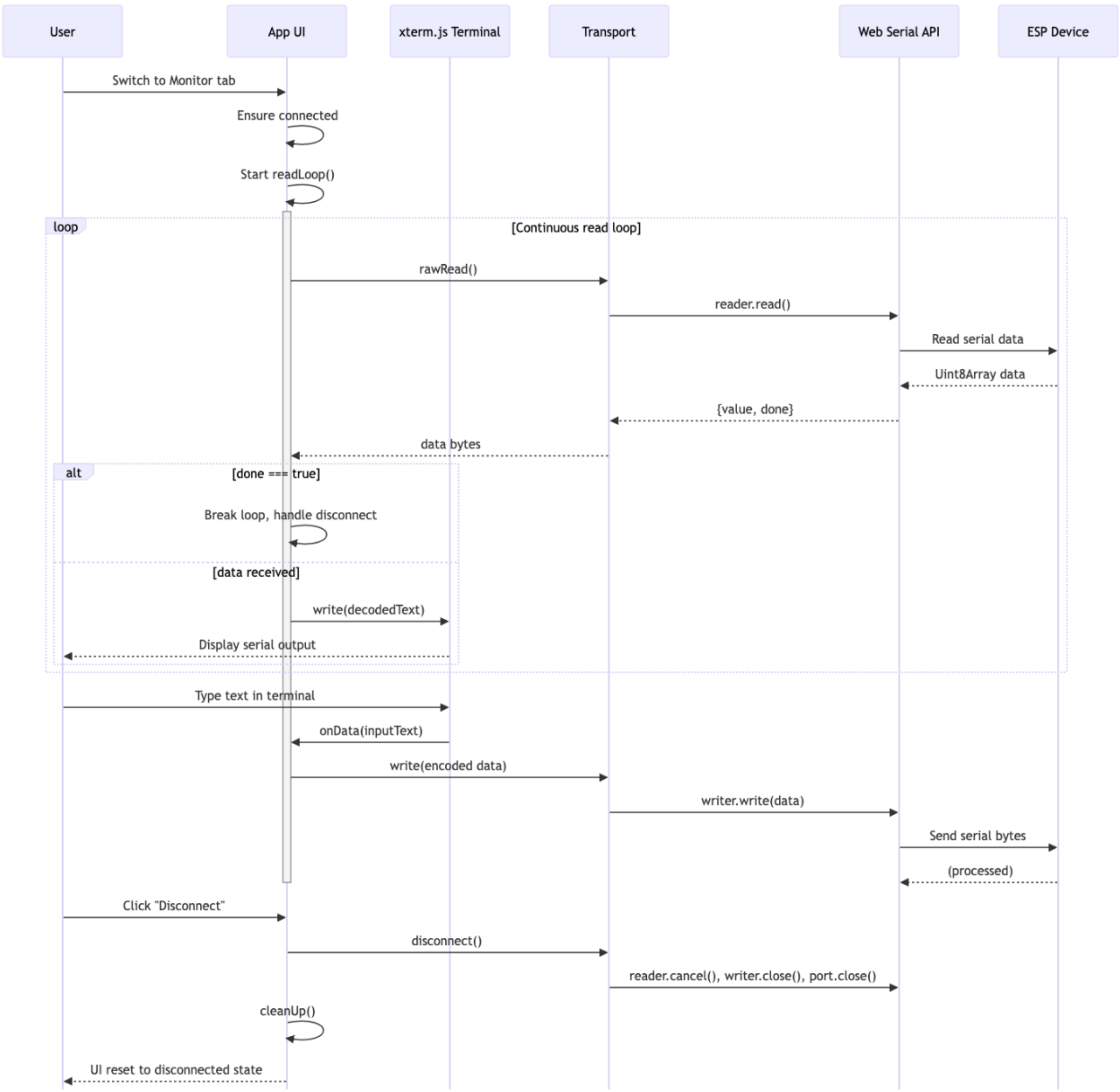
```
classDiagram
    class App {
        +HTMLElement terminal
        +HTMLElement fileArea
        +Transport transport
        +ESPLoader esploader
        +string chip
        +string device
        +number fileEntryId
        +boolean isConnected
        +boolean isFlashing
        +init()
        +btnConnect()
        +btnDisconnect()
        +btnProgram()
        +addFileEntry()
        +removeFileEntry()
        +readLoop()
        +sendData(data)
        +cleanUp()
    }

    class Transport {
        +SerialPort device
        +ReadableStreamDefaultReader reader
        +WritableStreamDefaultWriter writer
        +connect(baudRate)
        +disconnect()
        +read(timeout)
        +write(data)
        +rawRead()
        +setDTR(value)
        +setRTS(value)
    }

    class ESPLoader {
        +Transport transport
        +number baudrate
        +string chip
        +connect()
        +detectChip()
        +flashId()
        +eraseFlash()
        +writeFlash(fileArray, eraseAll, compress, onProgress)
        +changeBaud(newBaud)
        +hardReset()
        +softReset()
    }

    class Terminal {
        <<xterm.js>>
        +write(data)
        +clear()
        +onData(callback)
        +open(container)
    }
```

```
class CryptoJS {
    <<CDN Library>>
    +MD5(data)
}

class FileEntry {
    +number id
    +string filename
    +ArrayBuffer data
    +string offset
    +string md5Hash
}

App --> Transport : creates
App --> ESPLoader : creates
App --> Terminal : uses
App --> CryptoJS : hashes firmware
App --> FileEntry : manages 1..*
ESPLoader --> Transport : communicates via
Transport --> "Web Serial API" : uses
```

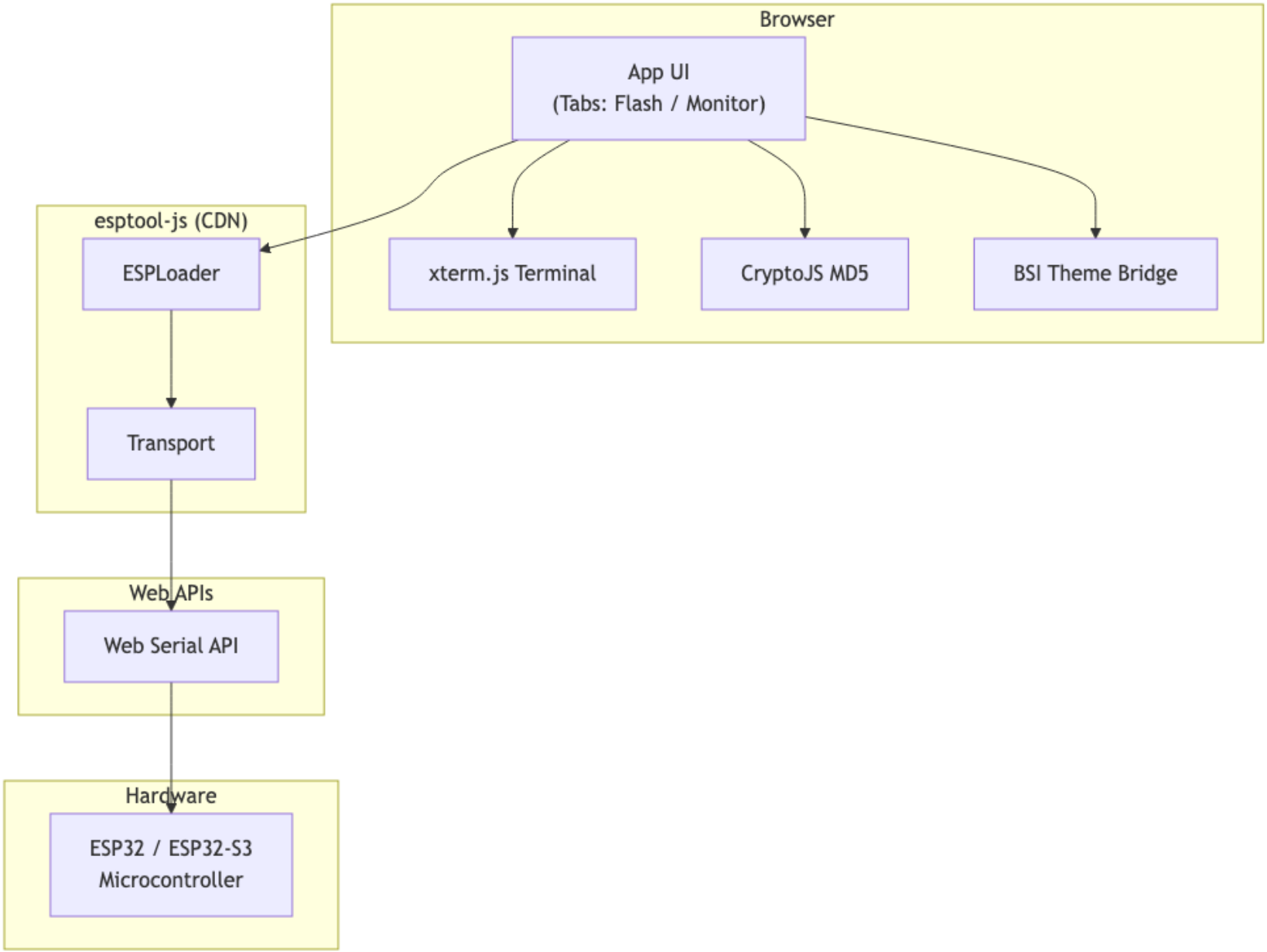
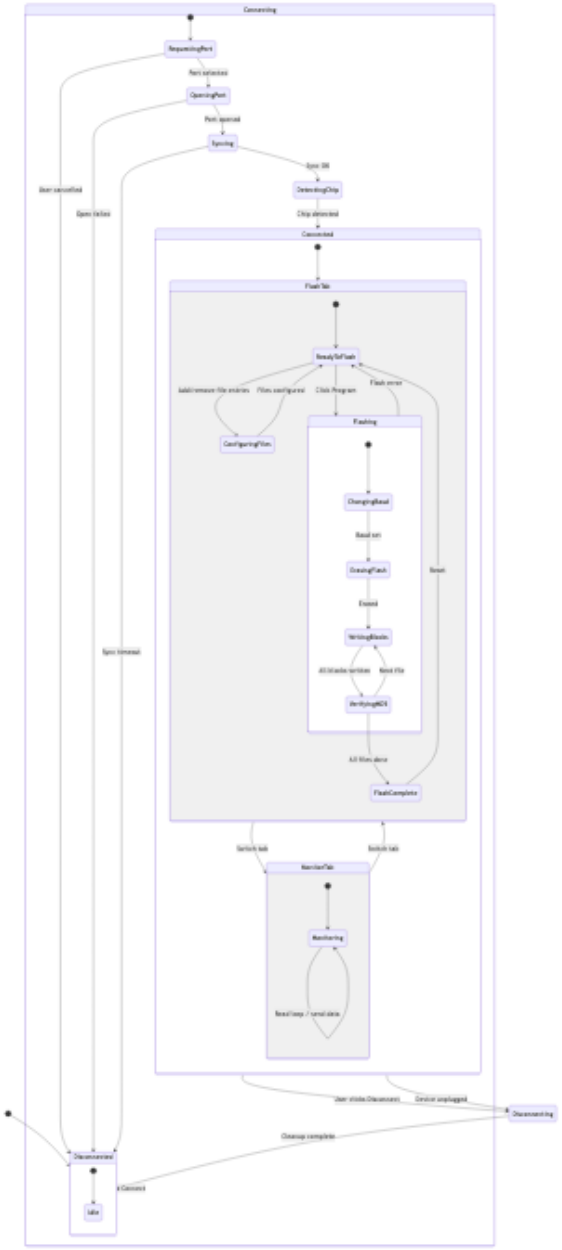
Sequence Diagram ? Flash Firmware Workflow



Sequence Diagram ? Serial Monitor Workflow



Component Diagram



State Diagram